



Characteristics of soil infiltration response to rainfall in differently converted forest hillslopes in a southern subtropical area of China

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Abstract

Purpose Understanding the characteristics of soil infiltration response to rainfall is critical for soil water transport processes and hydrologic modeling. However, little is known about how they vary with forest conversion at different temporal stages (< 20 years) below the rooting zone. Therefore, this study aims to investigate soil infiltration response patterns in different subtropical forest conversion hillslopes, including mature original forest (thick root), young secondary forest (middle root), and very young secondary grassland (fine root), and analyzed the effects of environmental factors on the dynamics of soil infiltration.

Materials and methods Several metrics were evaluated to characterize and quantify the nature of these responses by estimating changes in the soil water content, the response time difference between two adjacent soil depths, and infiltration velocities for 1144 infiltration events at 6 locations on the three hillslopes.

Results and discussion Soil infiltration responses were similar on both forestland hillslopes, yet significantly different from those on the grassland hillslope. The preferential flow was more evident in the profile of thick-rooted vegetation, and the velocity of the wetting front was faster in the profiles of middle- and fine-rooted vegetation. Topography and root characteristics interact to influence soil infiltration response at the hillslope scale.

Conclusions Conversion from thick-rooted forests into fine-rooted grasslands altered the rainfall-related soil infiltration dynamics below the rooting zone. In particular, the occurrence of preferential flow and infiltration rates varied, which helps enhance our understanding of ecohydrological processes in the context of changing land use and hydroclimatic conditions.

Keywords Soil infiltration response · Forest conversion · Hillslope hydrology · Humid area

1 Introduction

Soil infiltration is the process where water enters the soil from the surface (Zhang et al. 2017; Zhao et al. 2022). This process exerts a dominant control over the hydrologic cycle,

soil water storage, and plant-water relations (Cui et al. 2019; Guo and Lin 2016; Tromp-van Meerveld and McDonnell 2006). Generally, the forms of soil infiltration include matrix flow and preferential flow. Matrix flow is where water passes through the soil body and is a relatively slow, and even movement (Allaire et al. 2009). Whereas preferential flow is a fast form of soil water passing through macropores, in addition, it can accelerate deep drainage and/or water exchange (Beven and Germann 1982). In forest ecosystems, these two infiltrations are the critical “conveyor belt” to maintain the operation of various processes, such as connecting soil water storage, groundwater recharge, and plant water uptake (Dymond et al. 2021; Fuhrmann et al. 2019; Lin et al. 2006; Zhao and Wang 2021). Therefore, research on soil infiltration mechanisms is particularly important for forest water resource management and restoration of degraded vegetation.

The expansion of the agricultural frontier primarily drives deforestation, especially in one of the more common

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subtropical forest ecosystems. To maintain their cultivated areas, many native forests have been converted into farmlands (Cristiano et al. 2020; Zhang et al. 2022; Zhou et al. 2020). In this scenario, deforestation often affects the spatio-temporal distribution of rainfall through complex functional structures, such as the canopy, litterfall, and soil properties (Germer et al. 2010; Yu et al. 2018; Zhu et al. 2019). For example, Soares et al. (2021) observed that macroporosity in converted cropland soils was greatly reduced by compaction and that hydraulic conductivity in the upper soil layers could decrease to a certain extent. Accordingly, the increment or decrement in rainfall infiltration forms would show some differences in different vegetation units (Kurzweil et al. 2021; Litt et al. 2019; Zhang et al. 2023). On the other hand, the conversion of thick-rooted forests to fine-rooted croplands and/or grasslands leads to significant changes in root morphology, which further affect water infiltration through the living roots embedding, entangling, and enwrapping in soils (Liu et al. 2020b; Tian et al. 2017; Zhang et al. 2023). Thus, rainfall-related soil infiltration dynamics are worth exploring during forest conversion.

The interaction between vegetation and root can greatly change soil characteristics, as well as affect soil infiltration (Cui et al. 2019; Gerke et al. 2015; Jin et al. 2020; Liu et al. 2019). Benegas et al. (2014) noted that both live and rotting roots can positively affect soil permeability by forming root channels. Some studies showed that the macropores established through well-developed large root systems may cause rapid lateral and vertical flow of water infiltration as well as accelerate the preferential flow process after rainfall (He et al. 2012; Tian et al. 2019). Conversely, Li et al. (2021) concluded that fine and middle roots form the main channels that support preferential flow based on dye tracer experiments. In addition, other studies have also found similar patterns under different conditions (Jiang et al. 2018; Mawodza et al. 2021). Therefore, the infiltration characteristics of preferential flow after vegetation is combined with the conversion of root morphology remain unclear.

Deforestation also exerts different effects on different aspects of the soil infiltration response (e.g., response time, infiltration velocity, or changes in volumetric water content). Li et al. (2014) described a more dynamic soil infiltration response regime in cropland than in forestland after long-term forestland-to-farmland conversion in the southeastern subtropical area of China. Queiroz et al. (2020) found that long-term deforestation in the semiarid region of Brazil could decrease the volumetric moisture content in the upper soil layers. In addition, Zhao et al. (2013) reported a significant increase in soil infiltration rates over a large timescale during the restoration of deforested vegetation.

However, according to the infiltration patterns reported in these cleared cases, native forests were cleared between at least 30 years and even up to 100 years ago. Such estimated response patterns of soil infiltration provide limited information on the evolution of rainfall-related soil moisture dynamics after deforestation. Thus, the short-term temporal (< 20 years) characteristics of soil infiltration need to be studied at different conversion stages.

In situ real-time moisture monitoring at multiple depths is of great value for capturing the transient soil infiltration behavior after rainfall. Compared with traditional methods (e.g., double-ring infiltrometer, trace infiltration, automatic measurement system) (e.g., Flury and Flühler 1995; Liu et al. 2019; Zhao et al. 2022), this measurement displayed several advantages, including minimal disturbance to the original soil properties, allowing for repeatable observations at long-term scales. In addition, the capacity to capture high temporal-resolution dynamics of preferential flow processes on an event basis was displayed (Dymond et al. 2021; Kim et al. 2007; Lin and Zhou 2007; Lozano-Parra et al. 2015). For example, Lozano-Parra et al. (2015) used 30-min real-time moisture data over 2 years to identify the role of vegetation cover in the soil-wetting processes and found that vegetation was more decisive under drier environmental conditions. Wickenkamp et al. (2016) used 15-min soil moisture data for more than 3 years to identify the different types of soil moisture response to rainfall. They found that the spatial pattern of preferential flow in the profile was determined by microtopography, precipitation characteristics, and antecedent moisture conditions. Tian et al. (2019) collected real-time data to characterize soil infiltration dynamics from a long-term, large-scale real-time monitoring network with vegetation succession sequences in high and cold mountain areas. These studies demonstrate the potential of high-frequency moisture series for investigating the dynamic patterns of soil infiltration in response to rainfall.

In this study, we focused on the characteristics of the rainfall-related soil infiltration response processes below the rooting zone, as well as the relative influence factors at different forest-converted stages. Specifically, the objectives are as follows: (1) to quantify the change tendency of soil infiltration response patterns (e.g., response time and infiltration rate) along different forest conversion stages and (2) to evaluate the relative influencing factors of soil infiltration to provide suggestions for forest water resource management. We hypothesized that (1) the degree of soil infiltration response (e.g., response magnitude and response velocity) would increase with the restoration timescale of deforested vegetation, and (2) the vegetation types, root systems, and topographic characteristics interact to influence soil infiltration response.

2 Materials and methods

2.1 Study area

This study was conducted at the National Field Station for the Forest Ecosystem of Dalaoling, approximately 40 km northwest of the Three Gorges Dam in Yichang City, China (Fig. 1 and Table 1). This experiment focused on a small headwater catchment with the elevation ranging from 1100 to 1350 m, and average hillslope gradients from the channel to ridge varying from 10 to 55%. This area belongs to a typical subtropical monsoon humid climate zone with abundant precipitation; however, it is highly heterogeneous over the seasons. The 24-year mean annual precipitation measure near the field station from 1996 to 2019 was about 1400 mm, with 70% rainfall occurring in the growing seasons (May to

October). During the rainy season, a long drought period generally appears between July and September owing to strong evapotranspiration and low rainfall. The mean monthly temperature near the studying area averaged 31 °C in summer and 16.7 °C throughout the year. The potential evapotranspiration varies between 950 and 1200 mm/year (Liu et al. 2020a; Yi et al. 2019).

The soil of this study area is yellow–brown mountain soil according to international soil texture classification standards, and the soil thickness of the three hillslopes ranged from 60 to 125 cm depending on slope position. The soil texture of the six 0–40 cm soil profiles was mainly composed of silt and sand (Table 2). The volumetric rock fragment content (particles larger than 2 mm) is > 40% in the soil, resulting in the forestland having a high surface hydraulic conductivity. The bedrock is mainly composed of granite

Fig. 1 The location of the study area (a), vegetation types within the study hillslope (b), the terrain of the hillslope, and distribution of monitoring sites for soil moisture (c). MT, YM, and GF stand for mature forest, young forest, and very young grassland, respectively. U and D represent uphill and downhill, respectively. A diagram illustrating the process of thick-rooted forest converting into fine-rooted grassland (d), and details about the information of land use pattern can be seen in Table 1

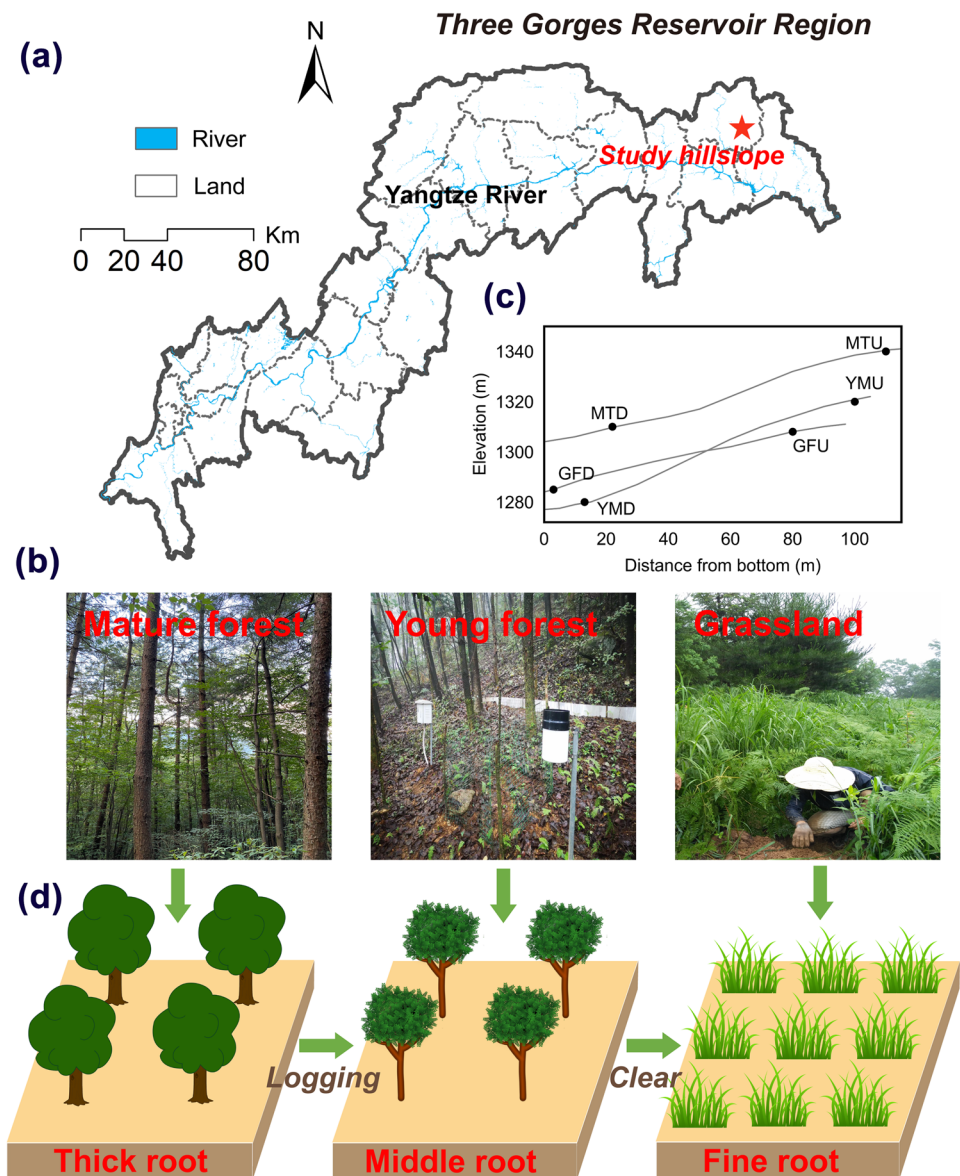


Table 1 Study hillslopes characteristics and land use histories

Hillslope	Land cover now	Land use history	Slope length
MT	Old secondary coniferous forest	40+ years old secondary growth/mature original forest	120 m
YM	Young secondary-growth forest	9–12 years old successional forest growing after logging or deforestation	115 m
GF	Secondary grass	Agriculture for 10 years old and later natural regrowth of native grass for ~2 years	107 m

rock, which is believed to be almost impermeable. The combination of high soil surface infiltration and low granite rock percolation rates led to a significant soil-rock interface, which triggered subsurface stormflow at the interface under heavy rainfall conditions (Zhang et al. 2021).

Rainfall is primarily routed to streams through surface and subsurface flows. Water moving along preferential paths through the soil profile is a particularly vital drainage mechanism in this area, with highly variable infiltration rates depending on the vegetation types. The vegetation in this area is dominated by second-growth broadleaf deciduous forests. Similar to the majority of this region, the old-growth coniferous forest at the field station has been clear-cut for the economic benefits (e.g., logging or conversion into farmland) since the 1970s. Three hillslopes of similar slopes, soil texture classes, and geology but with different vegetation cover and land-use history were selected (Table 1 and Fig. 1).

2.2 Data collection

Based on a detailed field investigation of the entire hillslope, two sites were selected for soil water content monitoring on each hillslope: one on the uphill slope and the other on the downhill slope (Fig. 1). In the summer of 2017, the basic characteristics (elevation, slope percent, depth to soil-saprolite boundary, and root range) and soil properties of three hillslopes were measured (Table 2). This experiment focused on the extensive rainfall periods that were likely to generate a detectable response; therefore, we analyzed soil moisture observations during the growing seasons (GY, from May to October of 2018–2019; Fig. 2).

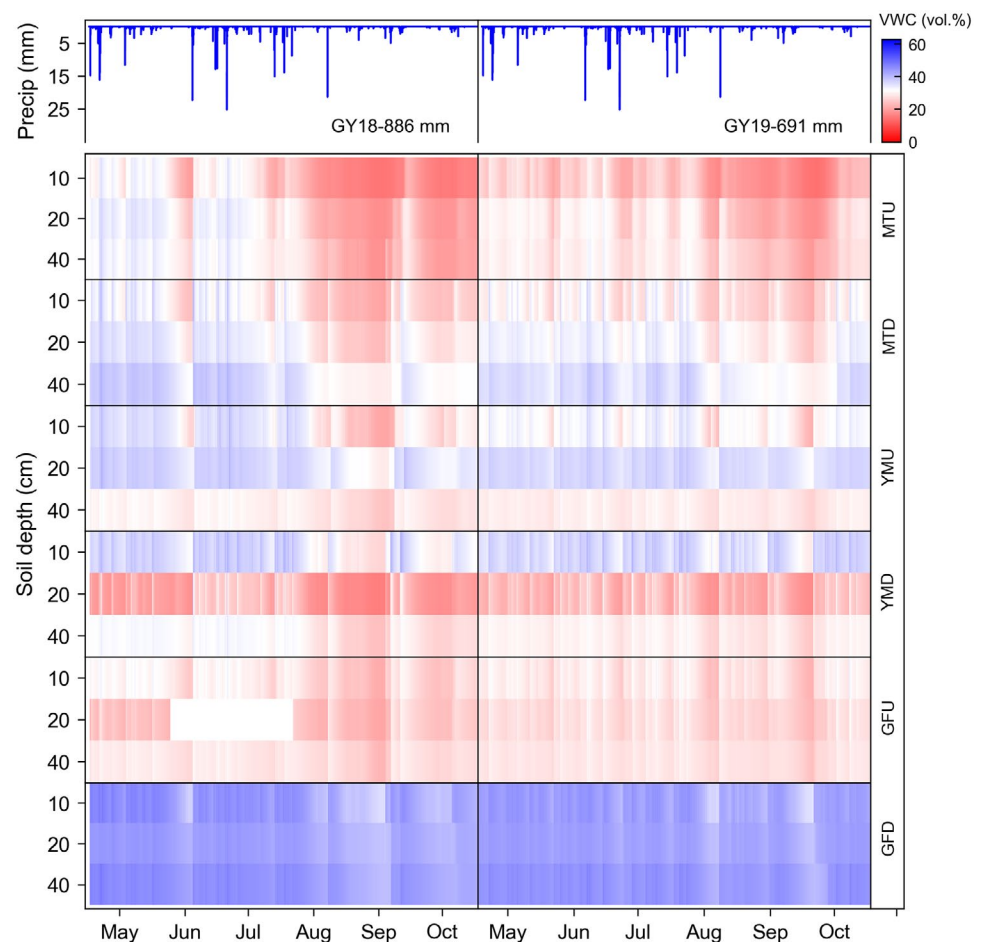
To quantify soil water dynamics, EC-5 probes (The METER Group, Inc., Pullman, Washington) were installed vertically in the soil at depths of 10, 20, and 40 cm. Probes were installed at both monitoring sites on each of the study hillslopes (6 sample sites, 18 total

Table 2 Soil properties and basic characteristics of the three study hillslope sites

Site	SD (cm)	Clay (%)	Silt (%)	Sand (%)	P (%)	SOM (g kg ⁻¹)	Ks (cm s ⁻¹)	BD (g cm ⁻³)	DB (cm)	RR (cm)	EL (m)	SP (%)
MTU	0–10	16.6	41.0	42.4	63.6	93.7	5.0E-03	0.76	90	95	1340	25
	10–20	15.9	48.1	36.0	62.7	64.7	2.7E-03	0.88	-	-	-	-
	20–40	13.2	46.2	40.6	55.9	46.6	1.5E-03	1.01	-	-	-	-
MTD	0–10	23.9	44.3	31.8	62.9	61.6	3.7E-03	0.66	120	90	1310	35
	10–20	26.5	48.5	25.0	61.3	48.1	2.2E-03	0.86	-	-	-	-
	20–40	27.8	46.7	25.5	57.7	27.4	8.7E-04	0.95	-	-	-	-
YMU	0–10	23.8	51.7	24.5	56.2	78.2	2.4E-03	1.08	90	75	1315	27
	10–20	38.1	47.4	14.5	52.8	23.4	1.6E-03	1.16	-	-	-	-
	20–40	25.6	31.2	43.2	47.5	14.2	8.7E-04	0.99	-	-	-	-
YMD	0–10	22.8	53.0	24.2	60.3	59.0	2.1E-03	0.72	125	70	1280	32
	10–20	24.1	53.8	22.1	59.1	51.3	1.4E-03	0.77	-	-	-	-
	20–40	32.1	41.1	26.9	52.7	14.0	3.3E-04	1.15	-	-	-	-
GFU	0–10	23.2	43.2	33.6	56.6	64.4	1.6E-03	0.78	85	35	1310	34
	10–20	16.7	46.2	37.1	55.3	50.2	1.0E-03	0.81	-	-	-	-
	20–40	18.6	43.2	38.2	52.2	35.0	6.5E-04	0.93	-	-	-	-
GFD	0–10	27.8	51.3	21.0	57.3	53.7	7.0E-04	0.69	60	32	1283	27
	10–20	27.8	56.8	15.4	55.3	43.3	4.8E-04	0.87	-	-	-	-
	20–40	29.9	46.1	24.0	54.2	11.4	1.7E-05	0.93	-	-	-	-

SD, soil depth, P soil porosity, SOM soil organic matter, BD bulk density, DB depth to soil-saprolite boundary, RR root range EL elevation, SP slope percent

Fig. 2 Distribution of precipitation and soil volumetric water content (VWC) during the growing seasons 2018 and 2019 for three depths (10, 20, and 40 cm) and six hillslope sites. Blank gaps indicate that data is missing here



probes) and connected to EM 50 data loggers (The METER Group, Inc., USA), which recorded data at 5-min interval. This study used data averaged from the original automatically monitored data to determine 30-min volumetric water content ($\text{m}^3 \text{m}^{-3}$).

Before the EC-5 probes were installed, they were calibrated for each site and soil depth in the field. Calibration was performed by combining gravimetric water content measurements with soil bulk density data for a range of volumetric soil water content under dry and wet conditions. During the installation, the probes were placed in the upslope direction to minimize the impact of disturbances on the water flow. Then, the soils were backfilled carefully to the previous bulk densities at corresponding depths. Finally, the first six month soil moisture data were excluded from the present study. Additionally, rainfall data were obtained at 30-min time intervals from a nearby automatic meteorological station (Fig. 2).

Before installing the monitoring systems, the soil profiles were opened and described; in addition, soil samples were collected at each moisture-monitoring site to measure key soil properties. Disturbed samples were air-dried at room temperature, mechanically disaggregated using a glass bar,

and dry-sieve using a 2-mm sieve. These soil samples were then used to measure volumetric particle size distribution (clay, silt, and sand content) and soil organic matter (SOM). The undisturbed core samples (100 cm^3 each) were used to quantify soil bulk density (BD), porosity (P), and hydraulic conductivity (K_s). The total porosity was then calculated using BD and soil particle density (2.65 g cm^{-3}).

2.3 Data analysis

In this study, the soil infiltration processes were divided into discrete infiltration events using methods similar described by Lozano-Parra et al. (2015) and Tian et al. (2019). The following criteria were applied for event selection: (i) a minimum threshold for the soil moisture response as an increase of 0.3 vol.% in 30 min, (ii) at least 6 h with no detectable soil moisture content increment as separated infiltration events, (iii) 30 min rainfall more than 0.2 mm during a 6-h period. In addition, 0.2 mm rainfall frequently occurs, and isolate records that did not produce any soil water content changed. For each of these events, a set of metrics was defined to characterize and quantify the soil water infiltration dynamics along the profile (Fig. 3).

Notably, the soil infiltration characterized in this study refers to total soil infiltration, including soil matrix infiltration and preferential flow.

The absolute change in soil water content (Δs , $\text{m}^3 \text{m}^{-3}$) was defined as the difference between the minimum and maximum moisture contents during each infiltration event. Based on the Δs , we further calculated the ratio (R_s) of Δs between the upper layers and adjacent lower layers for corresponding infiltration events. The absolute velocity (V_i , $\text{m}^3 \text{m}^{-3}/30\text{-min}$) of the rainwater infiltration was defined as the maximum slope of the percolation curve during the period of soil water increase. The start-to-start time (T_{s2s} , h) was defined as the time difference between the start responses of the upper and lower layers. Positive T_{s2s} values indicate that the moisture response of the upper layer occurred earlier than that of the lower layers, and negative T_{s2s} values indicate that the moisture response occurred earlier than that of the upper layers (suggesting the occurrence of preferential flow). The duration of the infiltration process (T_d , h) for a specific soil layer is defined as the time between the start and end time of the infiltration event. Finally, the absolute soil water storage increment (SWS) for each infiltration event was calculated using the following equation:

$$\text{SWS} = \sum \Delta s_z d_z$$

where SWS is the maximum soil water storage increment, and d_z is the thickness of each soil layer (i.e., 0–10, 10–20, and 20–40 cm depth).

2.4 Statistical analysis

The least significant difference (LSD) test was used to determine significant differences among the metric means (Δs , V_i , T_d) for different soil depths and land covers. A T -test was

performed to compare the differences in the metric means (R_s , T_{s2s}) between the two adjacent soil layers. Pearson's correlation was employed to test the influence of environmental variables on the analyzed response metrics. All statistical analyses were conducted in SPSS 19.0 (SPSS Inc., Chicago, IL, USA).

3 Results

3.1 Characteristics of infiltration events along the soil profile

Across the two growing seasons, the 1144 infiltration events identified can generally represent the wide range of rainfall infiltration mechanisms experienced in this area (Table 3). Approximately 42% were distributed in the first soil layer (L1: 0–10 cm), and 32% were distributed in the second layer (L2: 10–20 cm). Overall, the GF hillslope exhibited the highest percentage of infiltration events in L1, followed by YM, and the lowest in MT. However, the percentage of infiltration events in the deeper soil layer (L3: 20–40 cm) decreased in the following order: MT, YM, and GF, corresponding to the root range and diameter of the different vegetation. These results indicated that the decrease in infiltration events with depth may associate with the root characteristics of the vegetation. In other words, the deeper the root range, the thicker the root system, and the more infiltration events occurred in deeper soil layers.

3.2 Soil infiltration responses to rainfall

3.2.1 Response magnitude

As displayed in Figs. 4 and 5, the median Δs declined with soil depth, and R_s was typically above 100% in most

Fig. 3 Methodology used to identify the infiltration events. Vertical shaded areas represent the infiltration events occurred at the first soil layer in the profile

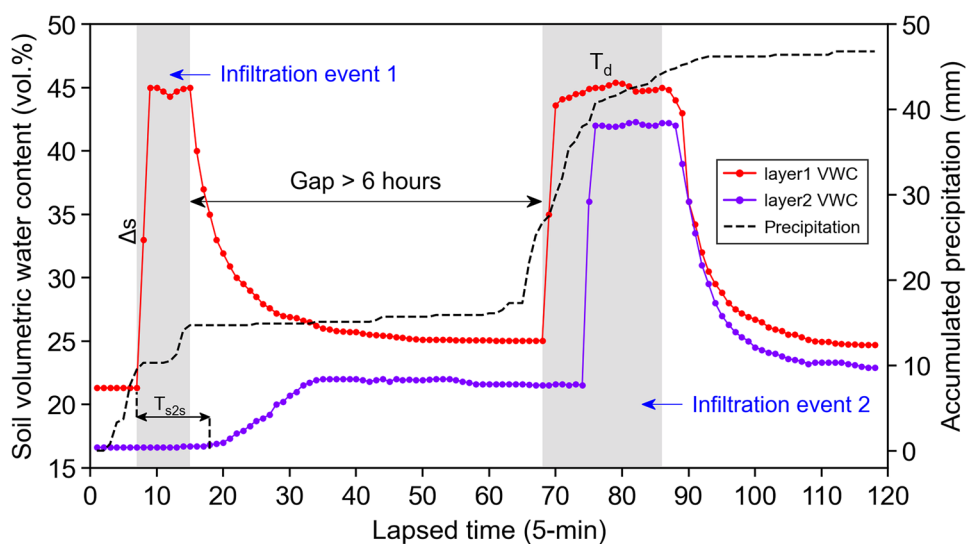


Table 3 Select numbers (percentage) of infiltration events in each soil layer in the monitoring sites

Soil layer	MTU	MTD	YMU	YMD	GFU	GFD	Sum
L1	55 (36.4)	83 (44.1)	82 (42.5)	96 (40.2)	83 (45.9)	83 (43.2)	482
L2	50 (33.1)	63 (33.5)	61 (31.6)	83 (34.7)	54 (29.8)	59 (30.7)	370
L3	46 (30.5)	42 (22.3)	50 (25.9)	60 (25.1)	44 (24.3)	50 (26.0)	292
Sum	151	188	193	239	181	192	1144

cases, suggesting that the magnitude of soil infiltration response to rainfall decreased with depth along the profile. Furthermore, median R_s declined with depth overall, indicating that the dampening patterns also decreased with increasing soil depth.

For MT, there was no significant difference in Δs and R_s along the profile at the uphill slope site, whereas median Δs and R_s exhibited an apparent decline at L2 and L3 at the downhill slope site. One possible explanation is that deeper soils displayed the lowest K_s (Table 2). In addition, compared with the other vegetation types, the MT hillslope had the fewest infiltration events in the soils (Table 3), and the temporal variation of water content was stronger in the shallow soil layer (Fig. 2).

The absolute change in soil water content (Δs) for YM displayed an abrupt increase from L1 (with a median of 3.7%) to L2 (with a median of 7.3%) at the downhill site (Fig. 4e), with the corresponding R_s below 100% at L2 (with a median of 59%) (Fig. 5e). In general, the lower the R_s , the higher the moisture content in the upper layer; however, this layer often has a lower average moisture content.

Our results also showed that the vertical distribution of the moisture content exhibited similar patterns at the YM downhill site (Fig. 2). Unlike the MT, median Δs in YM uphill slope site also had a significant decrease along the soil profile ($p < 0.05$) (Fig. 4b). Overall, compared with grassland hillslope (GF), the forest (MT and YM) downhill slope sites had a higher Δs in soils. Secondly, the forest hillslope showed a slightly higher R_s in the profile. These results indicated that hillslope covered by forest displayed a slightly dampened soil infiltration pattern.

For GF, median Δs showed a significant decline from L1 to L2 at both sites (Fig. 4c, f), and it was different from the observed pattern on the other hillslopes. In addition, the GF at the downhill slope site exhibited a notable sudden decrease in R_s with depth (Fig. 5f), whereas the moisture did not increase at L2 (Fig. 2). Also, the number of extreme values of Δs at this site was more frequent than that at the upslope site, further suggesting that the vertical variation of soil moisture content was not only affected by precipitation inputs, but also by other factors such as the fluctuation of ephemeral groundwater during rainfall events.

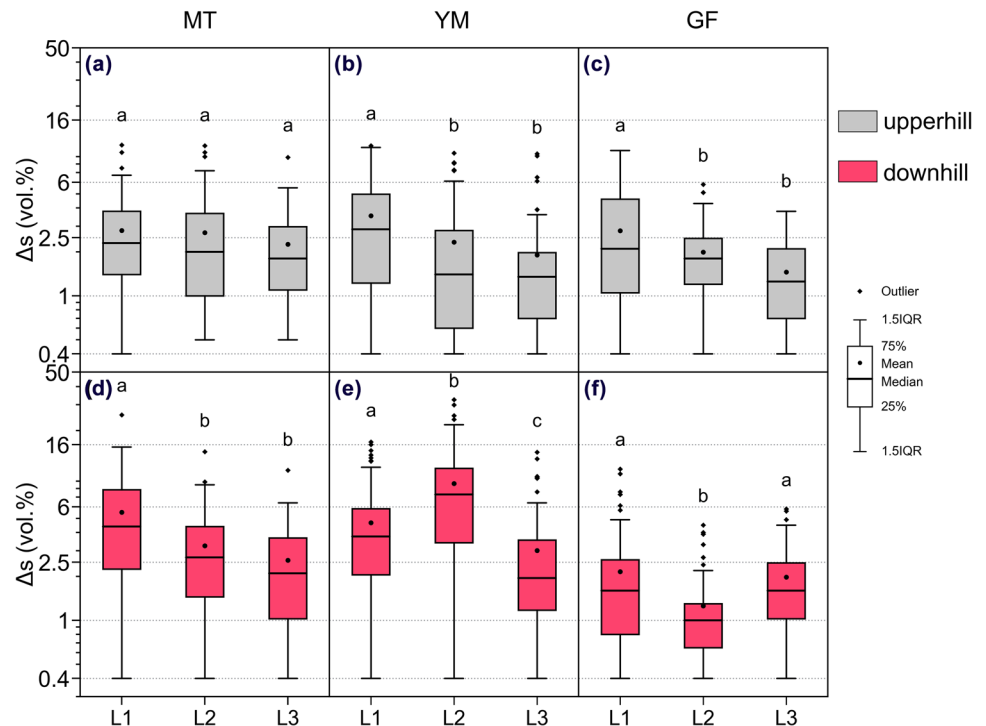
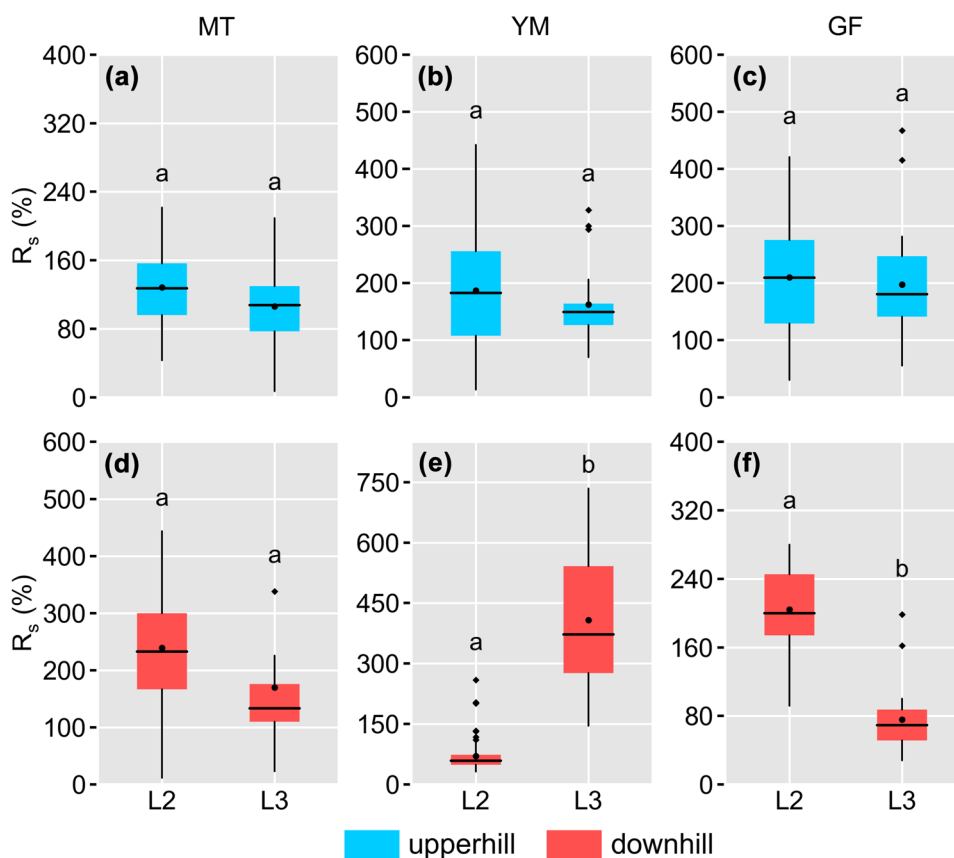
Fig. 4 Absolute changes in volumetric water content (Δs) for different monitoring sites and layers. The different lowercase letters above each box indicate the significant differences ($p < 0.05$) in different soil layers

Fig. 5 The ratio of Δs between two adjacent soil layers (R_s) for different monitoring sites and depths. The different lowercase letters indicate the significant differences ($p < 0.05$) in different soil layers



3.2.2 Infiltration velocity

Figure 6 shows the absolute velocity (V_i) distribution of the soil infiltration response along the profile for different vegetation types. Overall, median V_i decreased with depth in most cases, and the forest (MT and YM) downhill slope sites showed faster V_i at the corresponding depths; however, this was not observed at the grassland (GF) downhill slope site.

On the three hillslopes, YM and GF had a higher V_i along the soil profile, and no significant difference was detected in the median V_i across the two hillslopes ($p > 0.05$). Similar to the response magnitude, the median V_i increased significantly from L1 (median 4.8) to L2 (median 10.3) at the YM downhill slope site (Fig. 6e). Also, the V_i for the GF downhill slope site showed an increase from L2 (with a median of 0.08) to L3 (with a median of 1.8), which was significant at $p < 0.05$ (Fig. 6f).

3.2.3 Response time

To evaluate the susceptibility of soil infiltration under different land cover types, the start-to-start time (T_{s2s}) and duration of soil infiltration response (T_d) were extracted during each infiltration event in Figs. 7 and 8.

The T_{s2s} varied from negative values (indicating preferential flow) to a maximum of 49 h (GF), and its distribution patterns along the profile were very different from that of Δs and V_i . Overall, the median T_{s2s} increased following the order: MT, YM, and GF (Fig. 7). Furthermore, the median T_{s2s} at the downhill slope sites were often lower than those at the uphill slope sites.

On these three hillslopes, a large number of negative T_{s2s} values were observed on the MT hillslope (Fig. 7a, d), suggesting that preferential flow frequently occurred along the soil profile. In particular, the maximum T_{s2s} in the deeper soil layer (L3) reached a long time scale of 49 h on the GF hillslope, indicating that the downward transferability of water flow was poor in shallowly rooted soil. Moreover, the GF downhill slope site notably exhibits the lowest T_{s2s} at L3 (with a median of 0 h), and the range of T_{s2s} values varied from -6 to 49 along the whole profile. This is likely attributed to shallow groundwater facilitating rapid wetting of deep soils during rainfall periods.

The median T_d varied between 8.5 and 20 h for different vegetation types and hillslope positions (Fig. 8). Among them, the vertical distribution of T_d was relatively uniform along the YM profile, with a median ranging from 9 to 14 h at the uphill slope site, while ranging from 8.5 to 10.5 h at

Fig. 6 Absolute infiltration velocity (V_i) for different land covers and depths. The different lowercase letters denote significant differences ($p < 0.05$) in different soil layers

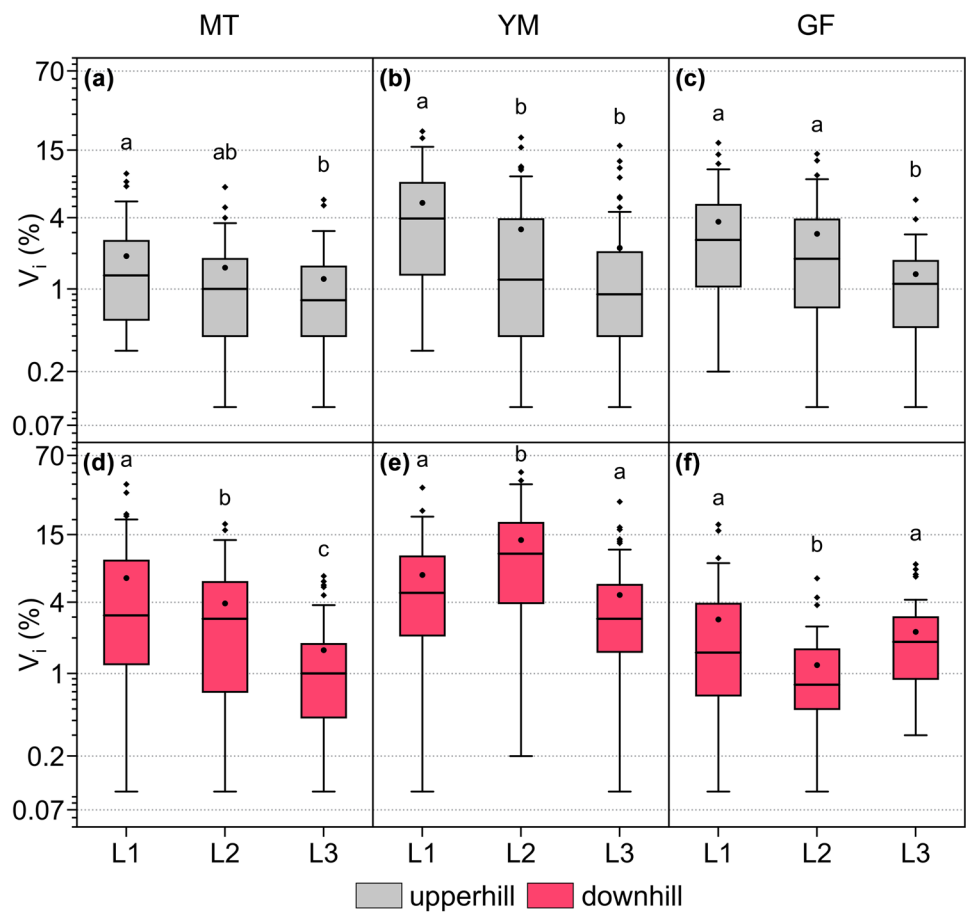


Fig. 7 The difference response time (T_{s2s}) for different monitoring sites and depths. The different lowercase letters above each box indicate the significant differences ($p < 0.05$) in different soil layers. The results of the T_{s2s} of all soil layers for different land covers (g). Inset in the right panel is the accumulated frequency of T_{s2s} for different land covers

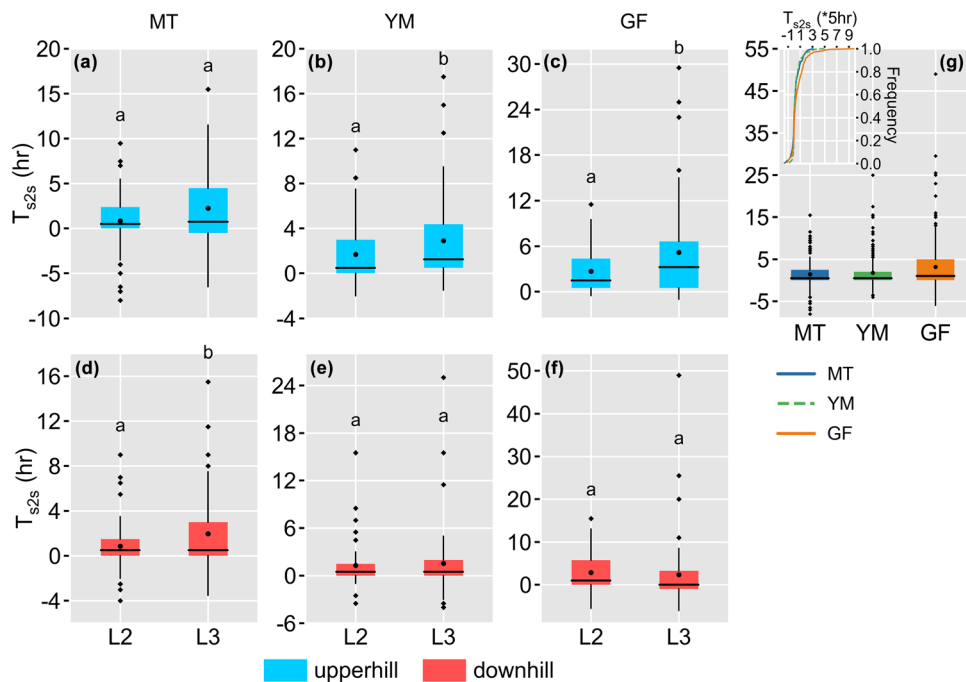
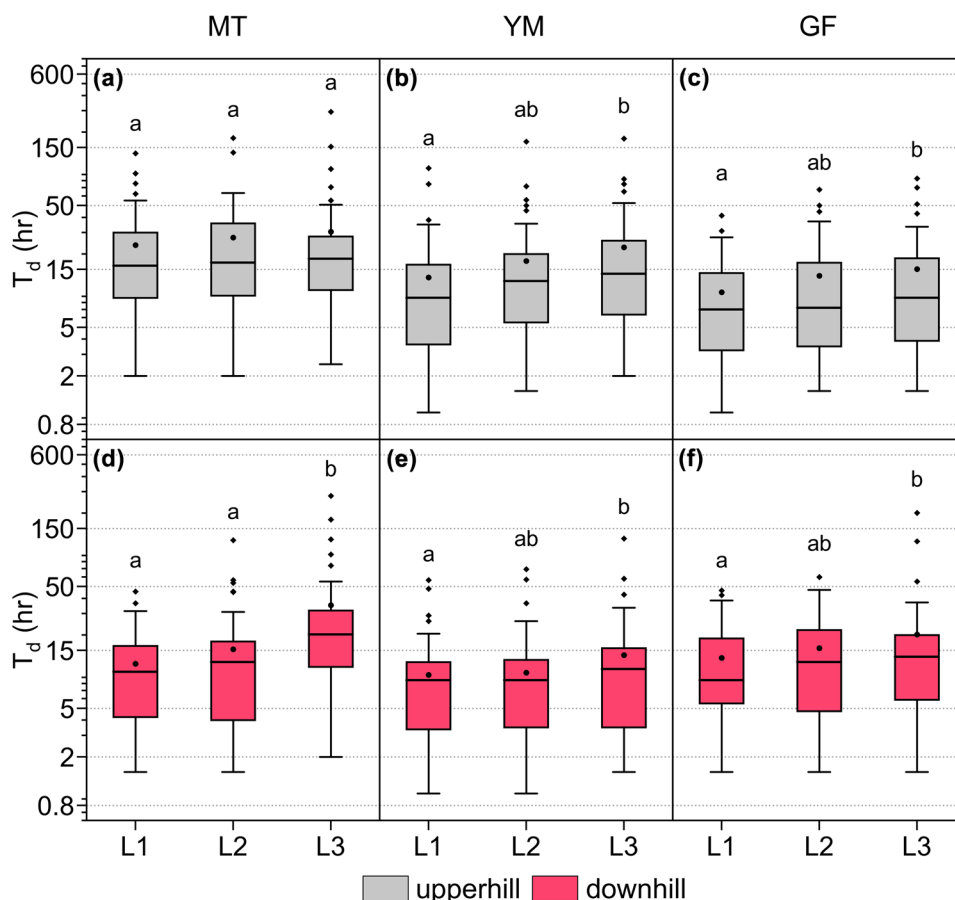


Fig. 8 Duration of infiltration events response (T_d) for different monitoring sites and depths. The different lowercase letters denote significant differences ($p < 0.05$) in different soil layers



the downhill slope site. The GF hillslope showed a similar pattern of T_d along the soil profile at the two hillslope positions, and the median T_d increased with depths from 7 to 13 h. In contrast to the pattern observed on the GF hillslope, the downhill slope site of MT exhibited a sharp increase from 10 h at L1 to 20 h at L3 ($p < 0.05$), with the maximum value reaching 200 h (about 8 days) (Fig. 8d).

3.3 Soil water storage along the profile

Figure 9 displays the distribution of soil water storage (SWS) and its proportion in each layer during the infiltration events. From the proportion of SWS (Fig. 9c, d), we can see that the first horizon (0–10 cm) generally showed the highest proportion of SWS. Among them, the highest and lowest averaged values were observed on the GF and MT hillslope, respectively. This result indicated that the MT hillslope displayed a more uniformly distributed SWS along the soil profile. For different hillslope positions, the accumulated proportion of SWS in the upper soils (0–20 cm) on the uphill slope increased in the order: MT (65%), YM (75.2%), and GF (77.9%). However, on the downhill slope, YM had the highest proportion (81.1%), followed by MT (80.2%), and GF showed the lowest proportion (71.8%). These results

indicate that the active soil depths varied with the vegetation conversion stages at different hillslope positions during infiltration events.

3.4 Relationships between the analyzed response metrics and environmental variables

On the three hillslopes, the relationships between response metrics analyzed (i.e., Δs , V_i , T_{s2s} , T_d , or SWS) and topographic factors (SP, EL, and DB) were the strongest on the YM hillslope, while weakest on the GF hillslope (Fig. 10b, c). Overall, the steeper the hillslope, the lower the site on the hillslope, such relationships were also more intense on average. However, a similar spatial variation was not observed on the GF hillslope (Fig. 10c), indicating that topography had less influence on soil infiltration responses on the GF hillslope. In addition, DB was positively related to Δs , V_i , and SWS, but negatively related to T_{s2s} and T_d .

The soil properties P and K_s were positively related to Δs and V_i ; however, they were negatively related to T_{s2s} and T_d (Fig. 10). The relationships between BD and response metrics were much stronger on both the MT and YM hillslopes, whereas such a relationship was relatively weaker on the

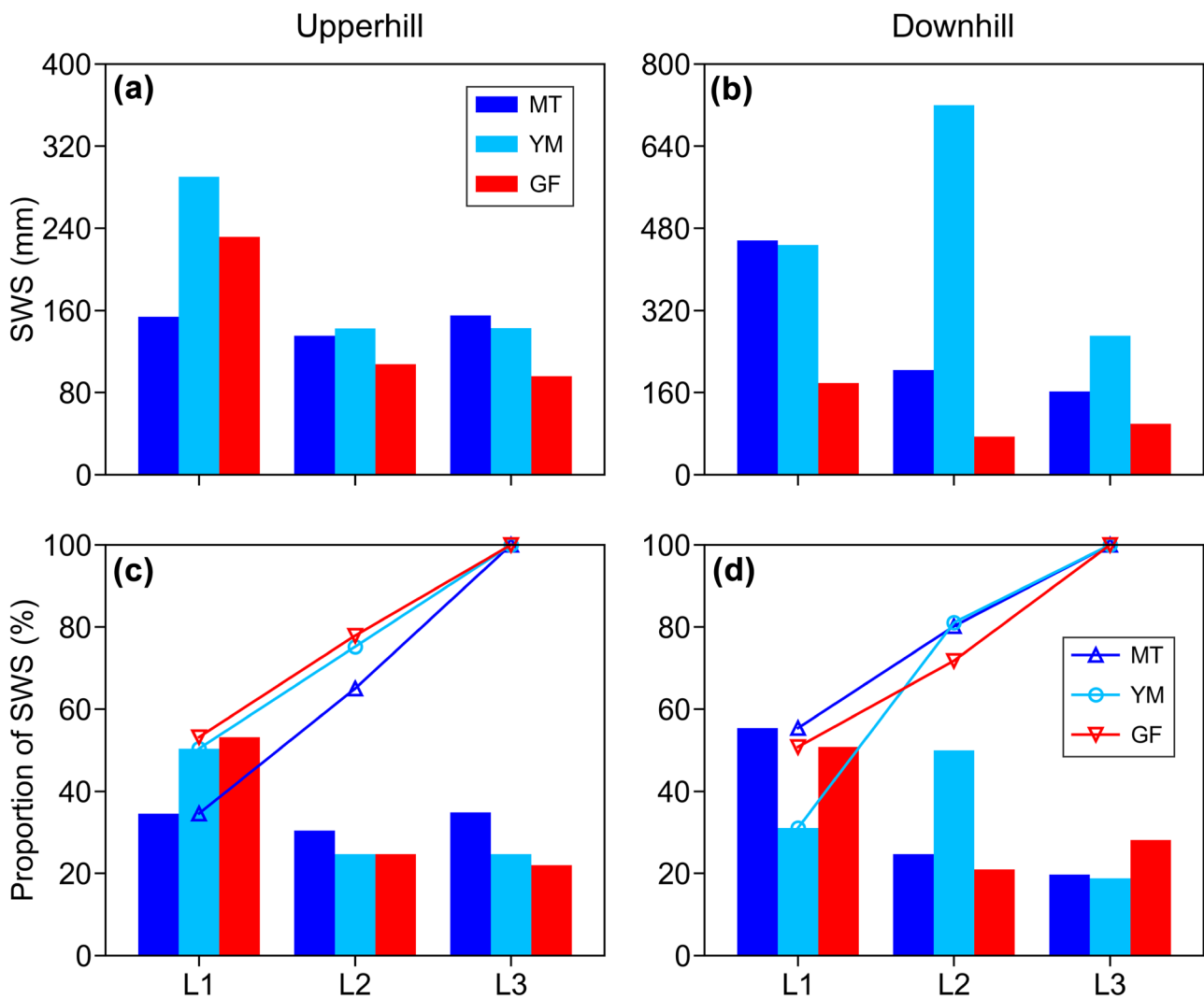


Fig. 9 Soil water storage increment (SWS) and proportion of SWS during the infiltration events for different land covers and soil layers

GF hillslope. Most of the infiltration metrics on the YM and GF hillslopes were negatively related to the silt and sand contents overall; however, such a relationship was not significantly related to the clay content (Fig. 10). In summary, a significant but relatively weak relationship exists between the infiltration metrics and the soil properties on the three hillslopes.

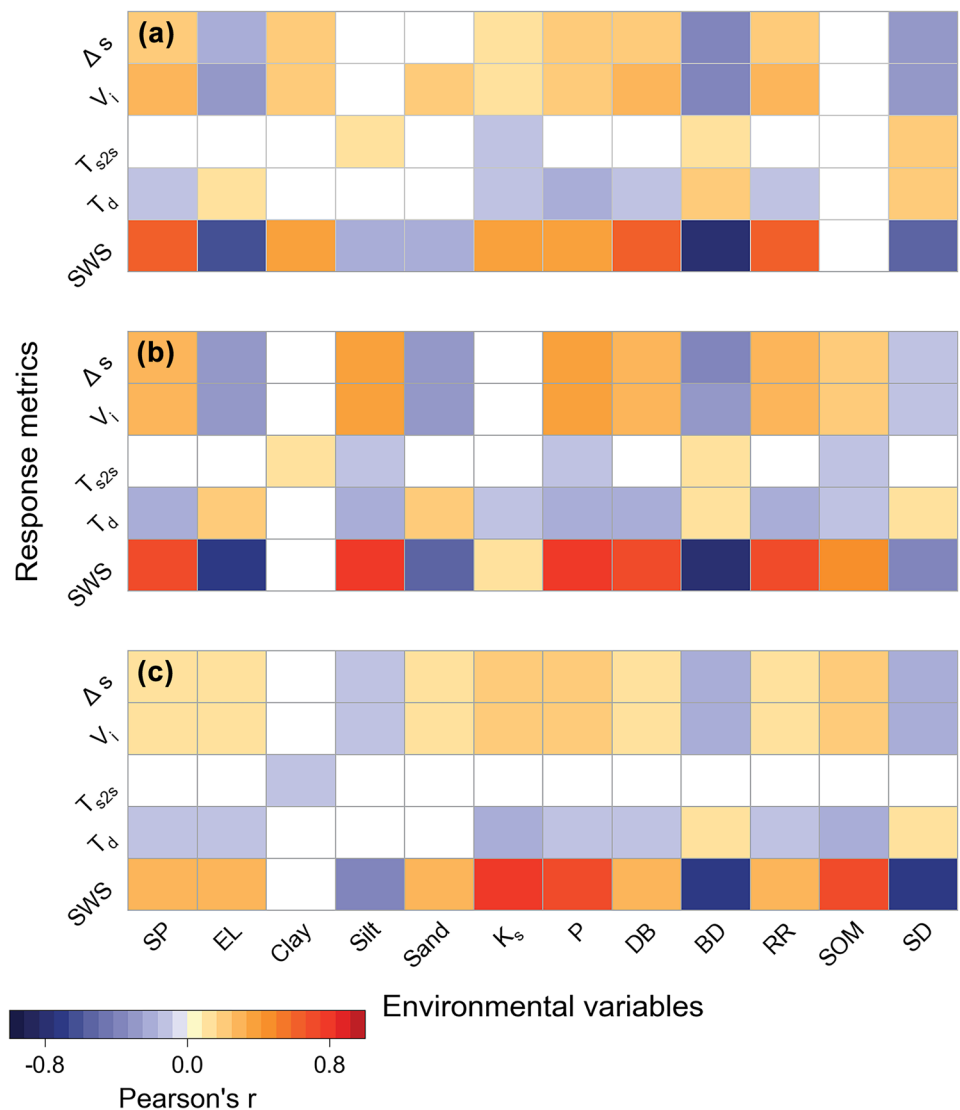
On both forest hillslopes, significant and strong relationships between RR and the response metrics were observed under most conditions (Fig. 10a, b). Among them, RR was positively related to Δs , V_i , and SWS, but negatively related to T_{s2s} and T_d . Furthermore, such a relationship was weakened or enhanced with the increasing SD (i.e., significantly negatively or positively related to SD). In other words, the combination of RR and SD regulated the dynamics of soil infiltration response to rainfall.

4 Discussion

4.1 Difference in soil infiltration dynamics for different land covers

Many studies demonstrated that land cover has a substantial impact on the hydrological processes of hillslopes, including soil infiltration, runoff generation, and nutrient cycling (Fan et al. 2016; Genxu et al. 2012; Jin et al. 2020; Zhu et al. 2014). However, details about the profile of the soil infiltration response of rainfall still require investigation in different reconverted forest hillslopes due to the complexity of root architecture and topography. To this end, we used a series of quantitative metrics to analyze the dynamic response patterns of profile soil infiltration under different vegetation-root-topography units based on infiltration events. In our

Fig. 10 Pearson correlation between the analyzed response metrics and environmental variables. **a–c** The MT hillslope, YM hillslope, and GF hillslope, respectively. Colored blocks represent significance at $p < 0.05$, and white blocks represent no significance



study, the moisture content varied strongly at both vertical and temporal scales on the MT hillslope (Fig. 2). Also, the fewest infiltration events and slowest infiltration velocity (V_i) were observed along the soil profile among all three hillslopes (Table 3 and Fig. 6). These findings indicated that effective rainwater infiltration could be controlled to a considerable extent by vegetation canopy interception intensity, thick litter layer, and water depletion. However, many negative T_{s2s} were in the profile, suggesting that a preferential flow frequently occurred on the MT hillslope (Fig. 7g). This result is consistent with the rainfall simulation experiments conducted at Agua Salud by Litt et al. (2019), which showed faster and deeper vertical transport in mature forest soils. Other studies have shown that forestland soils exhibit better permeability than grassland soils (Belsky et al. 1993; Li et al. 2021; Schaik 2009). Furthermore, in our study, significant Pearson's correlations between root range (RR) and response metrics were observed for both forestland

hillslopes (Fig. 10). The main explanation for this is that the MT soils have well-developed thick root systems, where water moves along the root-soil gaps through the profile allowing water to bypass part of the soil matrix (Hardie et al. 2011; Lin 2010).

On the three hillslopes, YM and GF exhibited higher infiltration velocities (V_i) along the soil profile (Fig. 6). In contrast to the thick roots that play an active role in forming preferred pathways and increasing rhizosphere soil particles (Gerke et al. 2015; Li et al. 2021), Nespoulous et al. (2019) demonstrated that live middle root, especially the fast-growing fine roots, could form a large number of non-capillary channels in a short time, thereby increasing soil infiltration capacity. Secondly, this was also attributed to the presence of many residual root macropores after logging or cultivation in shallow soils, which likely increased vertical hydraulic conductivity (Brouwer and Fitzpatrick 2002; Lin et al. 2006). In particular, coupled with the fact that the

degree of the YM root system developed between MT and GF, the active depth of subsurface flow was also shallower than MT in the profile (Allaire et al. 2009; Birch et al. 2021; Sidle et al. 2001), and thus, Δs and V_i were more affected by the lateral flow at the downhill slope site (Figs. 4 and 6).

On the GF hillslope, the mean T_{s2s} was longer than that on the other two hillslopes, and its maximum could reach a long time scale of 49 h in the deeper soil layer (Fig. 7f), which indicates that the downward transferability of water flow was poor in the shallowly rooted soil. This finding can also be seen from the lower values of Δs and V_i in the profile (Fig. 4c, f and Fig. 6c, f). In addition, owing to the differences in solar radiation, the moisture content exhibited substantial heterogeneity between the two hillslope positions (Fig. 2). However, the vertical distribution of the response metrics (i.e., Δs , V_i , and T_d) displayed a similar variation along the profile, indicating that soil infiltration in the downslope profile was less affected by subsurface lateral flow from the upslope area. These results suggest that the poorly developed root system eventually restrained water flow through the soil profile.

In summary, this study revealed that the dynamic response patterns of soil infiltration are considerably influenced by land cover and root systems. Compared with the grassland hillslope (GF), the forestland hillslopes (MT and YM) generally required a greater total rainfall amount to activate soil infiltration response, owing to the dense vegetation, high canopy closure, and thick litter layer. Secondly, the start-to-start time (T_{s2s}) on the forestland hillslope was shorter than that on the grassland hillslope, and extremely negative values were more frequent in the MT and YM soils, suggesting that the forestland profile had a slightly dampened effect on wetting front transport, and preferential flow occurred frequently along the soil profile. These results highlight the role of changing forestlands in regulating soil infiltration response to rainfall. Aboveground, canopy interception and stemflow altered the flux and spatial distribution of rainfall reaching the ground surface. During the rainwater infiltration period, the dynamic response patterns and frequency of preferential flow showed significant differences along the soil profile due to the different degrees of root system development.

4.2 Effects of soil properties on the soil infiltration

The mature forestland converted into young forestland or very young grassland might affect the soil properties on these three hillslopes, which could further impact different aspects of the soil infiltration response and soil water storage. Jia and Shao (2014) concluded that soil properties strongly affect dynamic response patterns of soil infiltration along the profile. The study by Lai et al. (2016) also shows that soil properties are crucial for regulating soil water movement rather than plant parameters based on field experimental methods and numerical simulation. In this study, the

relationship between BD and response metrics was much stronger on both forestland (MT and YM) hillslopes; silt and sand content were also found to be associated with these metrics on the YM and GF hillslopes. However, a significant but relatively weak relationship existed between the analyzed response metrics and soil properties overall (Fig. 10). On the one hand, due to the strong influence of control factors such as land cover, topography, and hydroclimate conditions, the impact of soil properties on the soil infiltration in this study was less profound. On the other hand, despite being reconverted into different land cover types in different years, no significant difference in the horizontal variation in soil texture was detected (Table 2). Therefore, soil texture did not lead to the larger variation in soil infiltration after land-cover change.

4.3 Effects of topography and shallow groundwater on the soil infiltration

In addition, topography also plays a critical role in regulating soil infiltration dynamics. On the forestland hillslope, compared with the uphill slope sites, the downhill slope sites generally showed a faster V_i , a shorter T_{s2s} , and a greater Δs at the corresponding depth (Figs. 4, 6, and 7). This result suggests that soil infiltration at the downhill slope site was more affected by subsurface lateral flow than that at the uphill slope site. Also, the vertical distribution of T_d showed that there might be an accumulation of soil water in the deep soils (40 cm) at the downhill slope site of MT (Fig. 8d), further suggesting that the subsurface lateral flow within the shallow soils from the upslope might increase soil water inputs at the lower slope site. However, no similar response patterns were observed for the GF hillslope. In a related study, Harmon et al. (2020) found that the mechanisms of vadose-zone root water uptake were significantly different on hillslopes: (a) upslope vegetation relies on soil water that is hydraulically disconnected from groundwater and (b) downslope vegetation relies on groundwater uptake instead. Our study also found that the SWS proportion within the top 20 cm of the upslope declined in the order of GF, YM, and MT (Fig. 9c), corresponding to the water depletion of vegetation. In contrast, on the downslope area, YM showed the highest proportion, followed by MT, and GF showed the lowest proportion (Fig. 9d), nearly corresponding to the root range of the different land covers.

Secondly, the fluctuation of shallow groundwater, combined with other factors in various ways, influenced the dynamic response patterns of deep infiltration. Although the groundwater table on these hillslopes was not directly investigated in this study, we could still deduce the potential influence of the groundwater table on soil infiltration by analyzing the deep moisture dynamics. In this study, because of the thin soil layer and poor permeability of granite coupled

with the near-stream location, the groundwater was relatively shallow at the GF downhill slope site. Therefore, the dynamics of soil infiltration response were affected more by the fluctuation of shallow groundwater. For example, during rainfall events, the shallow groundwater facilitated the rapid wet-up of deep soils, and thus, ephemeral saturation or near-saturation in deep soils (40 cm) occurred more frequently (Harmon et al. 2020; Tang et al. 2020), thereby significantly increased or decreased in Δs , V_i , and T_{s2s} (Figs. 4, 6, and 7). However, the depth of the groundwater table was still below that of the deepest sensor (40 cm).

5 Conclusions

In this study, a series of quantitative metrics was used to analyze the influence of thick-rooted mature forestland reconverted into middle-rooted young forestland and fine-rooted very young grassland on the soil infiltration response to rainfall, based on response magnitude, response velocity, and response time. Our results showed that the patterns of soil infiltration response were similar on both forestland hillslopes, yet significantly different from those on the grassland hillslope. In addition, the preferential flow was more evident in the thick-rooted vegetation profile, and middle- and fine-rooted vegetation profiles showed a faster response velocity of soil infiltration. Finally, topography and root characteristics interacted to influence the soil infiltration response at the hillslope scale. Our results provide new insights into the transient process of soil infiltration after the conversion of thickly rooted vegetation converts into medium- and fine-rooted vegetation. It provides valuable information for land use decision-making and physically based hydrologic modeling in the context of changing hydroclimatic conditions.

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Data availability Data used in this study are available upon request from the corresponding author.

Declarations

Conflict of interest The authors declare no competing interests.

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